Network Science (CSL601/CS552)

Mid-Semester Examination, IIT Bhilai
Exam Date: 19 Feb, 2025
Duration: 120 Minutes
Total Marks: 150

Instructions: Read the questions carefully before answering them. Return the question paper by attaching it along with the answer sheet.

Question 1: Multiple Choice Questions (MCQs)[18 Marks]

(a) Which of the following is the full form of GML?
   (a) Graph Modelling Format
   (b) Graph Manipulating Format
   (c) Graph Molding Format
   (d) Graph Making Format

(b) Which of the following formats is unsupported by Gephi?
   (a) GML
   (b) JSON
   (c) Pajek NET
   (d) Netdraw VNA

(c) If a node has a clustering coefficient 1, what is certain:
   (a) Degree Centrality is very high
   (b) Betweenness Centrality is very Low
   (c) None of its neighbors are connected
   (d) The node has only one neighbor

(d) In a star network, the central node has:
   (a) High betweenness and low closeness centrality
   (b) Low betweenness and high closeness centrality
   (c) High betweenness and high closeness centrality
   (d) Low betweenness and low closeness centrality

(e) Which of the following is a limitation of closeness centrality?
   (a) It works only in directed networks
   (b) It is undefined for disconnected networks
   (c) It may not be computed in weighted networks because the shortest path computation in weighted networks may be NP-Hard.
   (d) It is a local measure, i.e. is calculated based on the set of neighbors of a node.

(f) Which of the following statements are false in the context of the eigenvector centrality and its computation? [Multiple statements may be false. You must find all false statements to score the marks.]
   (a) Eigenvector centrality works only on directed networks.
   (b) It is a global measure, i.e., computing eigenvector centrality takes into account the whole network.
   (c) Starting with different initial vectors in the power method may lead to different rankings of nodes when the algorithm converges.

(d) Eigenvector centrality of a node is a function of the eigenvector centrality scores of the neighbors of that node.

Question 2: [20 Marks] Given a random network generated by Erdős-Rényi model with N nodes and the connection probability p.

(a) Calculate the probability that the generated network has L links.

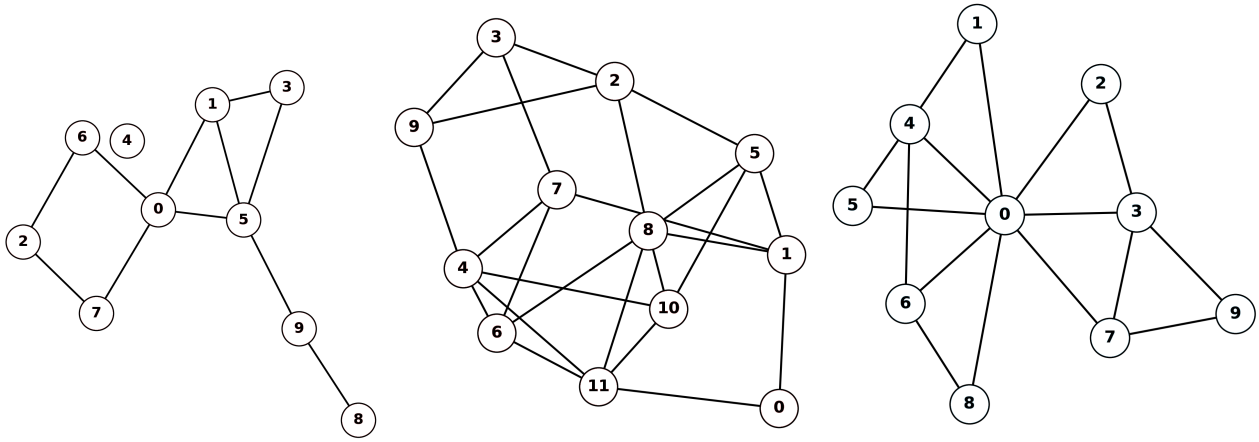
(b) Calculate the expected average degree.

(c) Calculate the probability that an arbitrarily chosen node has degree k.

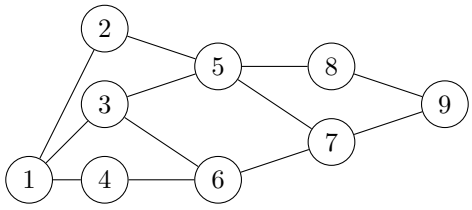
(d) Calculate the expected number of links.

Question 3:[5 Marks] Consider an undirected network with N nodes and $L = 3N$ edges. If a new node is added and connects to exactly half of the existing nodes, calculate the increase in the average degree.

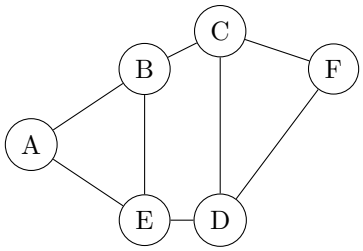
Question 4: [24 Marks] Three graphs are given below (numbered 1, 2, 3 from left to right). Draw the degree distribution histograms for each graph and identify the model that may have been used (ER or BA) to generate such a graph. Give one line justification in support of your identification. Estimate the p value if the Erdos-Rényi (ER) model is used or m value if Barabási-Albert (BA) model is used.



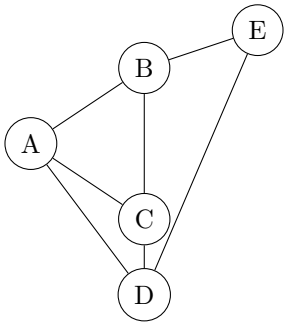
Question 5:[20 Marks] Consider the following network. Compute Graph Density, Diameter, Average Path Length, and Average Degree.



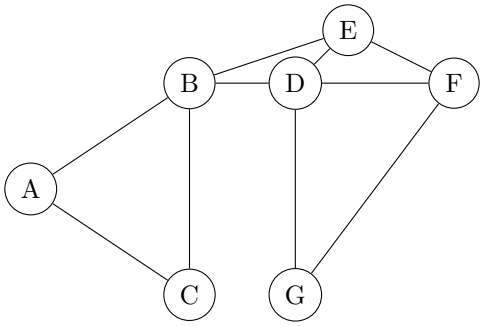
Question 6:[20 Marks] Compute the betweenness centrality using Brandes' Algorithm for the given graph. State the theorem given by Brandes which was used by him in his algorithm to reduce the time complexity of betweenness computation in sparse graphs. Give the proof idea.



Question 7:[18 Marks] Compute the eigenvector centrality using the power method for the following network. Start with a unit vector i.e., an initial vector containing all the entries as one. The vector is normalized after each iteration (before being used in the next iteration) using the $l^2 - norm$ of that vector. $l^2 - norm$ of a vector $[x_1, x_2, \cdots, x_n]$ is $\sqrt{\sum_{i=1}^{n}(x_i)^2}$. You must run the power method for 4 iterations.



Question 8: [15 Marks] Consider the following network. Calculate Degree Centrality, Betweenness Centrality, Closeness Centrality, Eccentricity, Harmonic Centrality, Clustering Coefficient for node A and node F.



Question 9: [10 Marks] Comment about the node's position in a network when

1. The node has low betweenness centrality but high closeness centrality.
2. The node has low betweenness centrality but high degree centrality.
3. The node has high betweenness centrality but low degree centrality.
4. The node has low closeness centrality but high degree centrality.